

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 02: CELL: THE BUILDING BLOCK OF LIFE

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
5. **Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): The plasma membrane is referred to as a selectively permeable membrane.

Reason (R): It permits the entry and exit of certain specific substances while preventing the passage of other materials across the cell.

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): Plant cells can withstand much greater changes in the surrounding hypotonic medium than animal cells without bursting.

Reason (R): Plant cells possess a rigid, tensile cell wall made of cellulose that exerts counter wall pressure against incoming turgor pressure.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.3

Assertion (A): Mitochondria are often designated as the 'powerhouses of the cell'.

Reason (R): Mitochondria synthesize adenosine triphosphate (ATP), the universal chemical energy currency required for cellular metabolic activities.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.4

Assertion (A): Lysosomes are commonly known as the 'suicide bags' of the cell.

Reason (R): When a cell suffers irreparable metabolic damage or dies, its lysosomes rupture, releasing powerful hydrolytic enzymes that digest the host cell.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.5

Assertion (A): Ribosomes are found in both prokaryotic and eukaryotic cells.

Reason (R): Ribosomes are membrane-bound organelles containing hydrolytic digestive enzymes.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.6

Assertion (A): Mitochondria and chloroplasts are considered semi-autonomous organelles.

Reason (R): They contain their own circular DNA and 70S ribosomes, enabling them to synthesize some of their own structural proteins and replicate independently.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.7

Assertion (A): Chromosomes are visible as distinct rod-shaped structures only when a cell is actively preparing to divide.

Reason (R): In a non-dividing cell, genetic material exists in the form of an uncoiled, entangled thread-like network called chromatin.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.8

Assertion (A): The inner mitochondrial membrane is extensively folded into finger-like projections called cristae.

Reason (R): Cristae substantially increase the total surface area available for ATP-generating respiratory enzyme complexes.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.9

Assertion (A): Prokaryotic cells lack a defined, membrane-bound nuclear region.

Reason (R): The undefined nuclear region containing naked nucleic acids in prokaryotes is designated as a nucleoid.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.10

Assertion (A): Smooth Endoplasmic Reticulum (SER) plays a crucial role in detoxifying many poisons and drugs in liver cells of vertebrates.

Reason (R): SER contains enzyme complexes that chemically convert lipid-soluble toxins into water-soluble compounds for excretion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)

[8 MARKS]

Q.11

Assertion (A): When dry raisins are soaked in pure water, they swell up after several hours.

Reason (R): Water enters the raisin cells via endosmosis because the external pure water is hypotonic relative to the concentrated cell sap.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): When swollen raisins are subsequently transferred into a concentrated salt solution, they shrink.

Reason (R): Water molecules move out of the raisin cells into the hypertonic external medium via exosmosis.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): A peel of *Rhoeo* leaf mounted in strong sugar solution displays plasmolysis under a microscope.

Reason (R): Severe loss of water via exosmosis causes the plant protoplast to shrink and pull away from the rigid cellulose cell wall.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): Boiled *Rhoeo* leaf cells do not undergo plasmolysis when placed in a concentrated sugar solution.

Reason (R): Boiling kills the plant cells, destroying the semi-permeable properties of the cell membrane and denaturing transport proteins.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): Red blood cells (RBCs) swell and burst (lyse) when placed in a hypotonic salt solution.

Reason (R): Animal cells lack a rigid cell wall to withstand increasing hydrostatic internal turgor pressure.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): Amoeba acquires its food from the external aquatic environment through the process of endocytosis.

Reason (R): The plasma membrane of Amoeba is flexible and dynamic, allowing it to engulf external food particles by forming pseudopodia.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): The Golgi apparatus is directly involved in the synthesis of structural proteins from amino acids.

Reason (R): Golgi apparatus consists of membrane-bound parallel stacks of flattened cisternae.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): Chloroplasts are known as the kitchen of plant cells.

Reason (R): Chloroplasts contain chlorophyll pigments and photosynthetic machinery that synthesize glucose using sunlight, water, and CO_2 .

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)**[7 MARKS]****Q.19**

Assertion (A): Endoplasmic Reticulum and Golgi apparatus operate in functional coordination within the eukaryotic endomembrane system.

Reason (R): Proteins and lipids synthesized in the ER are transported to the Golgi apparatus for chemical modification, packaging, and dispatch.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): Leucoplasts are primary sites for photosynthesis in plant stems.

Reason (R): Leucoplasts are specialized colorless plastids that store starch, oils, and protein granules in non-photosynthetic plant tissues.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): Large central vacuoles occupy 50% to 90% of the volume of mature plant cells.

Reason (R): The central vacuole maintains turgidity and rigidity in plant cells and serves as a reservoir for cellular waste and nutrients.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): Diffusion is an active transport process that requires continuous breakdown of cellular ATP molecules.

Reason (R): In diffusion, molecules move spontaneously from a region of higher concentration to a region of lower concentration along a gradient.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): The cell is recognized as the structural and functional unit of all living organisms.

Reason (R): All basic metabolic processes of life (respiration, nutrition, excretion) can be independently performed by a single individual cell.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): Nuclear membrane contains microscopic pores called nuclear pores.

Reason (R): Nuclear pores facilitate regulated bidirectional exchange of RNA and proteins between the nucleoplasm and cytoplasm.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): Viruses do not show characteristics of life until they enter a living host cell.

Reason (R): Viruses lack cellular machinery and membranes of their own and utilize the host cell's metabolic machinery to multiply.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01 ☐ A ☐ B ☐ C ☐ D	02 ☐ A ☐ B ☐ C ☐ D	03 ☐ A ☐ B ☐ C ☐ D	04 ☐ A ☐ B ☐ C ☐ D	05 ☐ A ☐ B ☐ C ☐ D
06 ☐ A ☐ B ☐ C ☐ D	07 ☐ A ☐ B ☐ C ☐ D	08 ☐ A ☐ B ☐ C ☐ D	09 ☐ A ☐ B ☐ C ☐ D	10 ☐ A ☐ B ☐ C ☐ D
11 ☐ A ☐ B ☐ C ☐ D	12 ☐ A ☐ B ☐ C ☐ D	13 ☐ A ☐ B ☐ C ☐ D	14 ☐ A ☐ B ☐ C ☐ D	15 ☐ A ☐ B ☐ C ☐ D
16 ☐ A ☐ B ☐ C ☐ D	17 ☐ A ☐ B ☐ C ☐ D	18 ☐ A ☐ B ☐ C ☐ D	19 ☐ A ☐ B ☐ C ☐ D	20 ☐ A ☐ B ☐ C ☐ D
21 ☐ A ☐ B ☐ C ☐ D	22 ☐ A ☐ B ☐ C ☐ D	23 ☐ A ☐ B ☐ C ☐ D	24 ☐ A ☐ B ☐ C ☐ D	25 ☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(A)	Assertion is true. Reason is true and directly defines the biological property of selective permeability.
Q.2	(A)	Assertion is true. Reason is true and explains how the cellulosic cell wall prevents osmotic lysis in plant cells.
Q.3	(A)	Assertion is true. Reason is true and correctly explains the biochemical origin of the nickname 'powerhouse'.
Q.4	(A)	Assertion is true. Reason is true and describes autolysis, which justifies the term 'suicide bags'.
Q.5	(C)	Assertion is true. Reason is false: Ribosomes are non-membrane-bound ribonucleoprotein complexes that synthesize proteins, not hydrolytic enzymes.
Q.6	(A)	Assertion is true. Reason is true and directly justifies why they are designated as semi-autonomous endosymbionts.
Q.7	(A)	Assertion is true. Reason is true and explains how chromatin condenses into distinct chromosomes during cell division.
Q.8	(A)	Assertion is true. Reason is true and provides the biological advantage of folded cristae in cellular respiration.
Q.9	(B)	Both statements are factually true, but Reason names the structure rather than explaining the evolutionary/structural cause of why prokaryotes lack a nuclear envelope.
Q.10	(A)	Assertion is true. Reason is true and explains the enzymatic mechanism of hepatic detoxification by SER.
Q.11	(A)	Assertion is true. Reason is true and correctly identifies endosmosis driven by a hypotonic water concentration gradient.
Q.12	(A)	Assertion is true. Reason is true and explains exosmosis occurring across the semi-permeable membrane into a hypertonic solution.
Q.13	(A)	Assertion is true. Reason is true and correctly describes the cytological mechanism of plasmolysis.
Q.14	(A)	Assertion is true. Reason is true and explains why dead cells cannot undergo selective osmosis.
Q.15	(A)	Assertion is true. Reason is true and explains why RBCs burst while walled plant cells remain intact.
Q.16	(A)	Assertion is true. Reason is true and explains how plasma membrane fluidity enables phagocytosis/endocytosis.
Q.17	(D)	Assertion is false: Ribosomes synthesize proteins, not Golgi (Golgi packages and modifies them). Reason is true (cisternae description is accurate).
Q.18	(A)	Assertion is true. Reason is true and directly explains why chloroplasts are termed the cell's 'kitchen'.
Q.19	(A)	Assertion is true. Reason is true and details the integrated secretory pathway of the eukaryotic endomembrane system.
Q.20	(D)	Assertion is false: Chloroplasts perform photosynthesis, leucoplasts do not. Reason is true: Leucoplasts are storage plastids.
Q.21	(A)	Assertion is true. Reason is true and explains the structural and physiological significance of the large central vacuole.
Q.22	(D)	Assertion is false: Diffusion is passive transport and requires zero ATP. Reason is true: It follows a spontaneous concentration gradient.
Q.23	(A)	Assertion is true. Reason is true and provides the biological justification for the cell theory postulate.
Q.24	(A)	Assertion is true. Reason is true and explains the physiological function of nuclear pore complexes.
Q.25	(A)	Assertion is true. Reason is true and explains why viruses are classified on the borderline of living and non-living.